

GYANQUEST

Project Report

Interactive bilingual mission games for school STEAM · Canvas 2D curriculum · 3D Specimen Lab

Offline-first · English + Bangla · Browser, not a hardware lab

Competition / event	BrainChild 2.0
Team name	Team Alpha
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No.	Full name	Role
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Declaration

We, the members of Team Alpha, declare that this report describes original work on GyanQuest. Cited ideas from education research are referenced. The software, curriculum design, and this document are submitted for the competition named above. We confirm that the product can be demonstrated in a browser.

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1. Abstract

GyanQuest is a bilingual (English and Bangla), offline-first learning platform that turns school STEM topics into playable Canvas 2D missions. One shared engine serves 28 subject titles. Learners act on a stage (drag, tap, predict, myth-bust), get live coach feedback, unlock a digital book, and may ask an optional AI tutor. Progress is gated by an 80% fluency check before Mastery. A separate 3D Specimen Lab lets learners inspect scientific models without forcing 3D onto every lesson. Core play works in a browser with no install and no cloud requirement. Optional Groq chat and a Laravel admin layer add accounts and tutoring when a network exists. Impact is converting curriculum into mastery-gated missions at the cost of a browser, not a hardware lab.

Indicator	Value (August 2026 live catalog)
Subject titles in the catalog	28 (all live folders)
Uniqueness-pass games	16
Finished missions	52
Instructional layers	4 (canvas, coach, book, AI)
Fluency gate before Mastery	80%
Language modes	2 (English, Bangla)

2. Introduction and problem

School STEAM in many Bangladeshi classrooms is still lecture-heavy. Tools are fragmented: a video here, a quiz app there, a 3D demo that only runs on a lab PC. Shared phones, flaky internet, and no dedicated hardware lab are the real conditions. Learners watch and click multiple-choice items. Teachers cannot see whether a concept was enacted or only memorized.

GyanQuest attacks that gap. The product is one engine, many subjects, short missions, and a fluency lock. It is designed so a village computer class and a city coaching center can run the same titles.

3. Objectives

- Turn stated learning objectives into playable missions with a clear win condition.
- Keep the play stage readable: Canvas 2D for graded missions; 3D only where spatial inspection helps.
- Restore hub vs in-mission state after reload (save-v2 in the browser).
- Offer Bangla and English as first-class modes, not an afterthought overlay.
- Keep AI optional: the game must teach if Groq is down or the key is missing.
- Stay install-free: a modern browser is the only required hardware.

4. Users and Bangladesh context

Stakeholder	What changes
Learner (roughly ages 9-16)	Enacts the idea, fails safely, retries; language toggle EN/BN.
Teacher / facilitator	One pattern across 28 titles; coach lines and books, not 28 different apps.
School / coaching center	No Unity install, no GPU lab; shared devices still work.
Competition jury	A runnable demo: landing, a 2D mission, Specimen Lab, optional tutor.

5. Solution overview

The learner path is: Landing catalog → subject game → Mission Hub → Canvas 2D mission → coach feedback → digital book → optional Groq tutor. Progress is stored locally. Login and admin sync exist but are optional.

Four instructional layers (canvas leads; everything else follows)

Layer	Job	Pedagogy
1. Interactive canvas	Source of truth. Drag, tap, tune, myth-bust.	Constructivism; Predict / Observe / Explain
2. Game coach	Live commentary on streaks and misses.	Formative assessment; Flow; ZPD scaffolding
3. Digital book	Retrieval practice and dual coding; term links.	Bloom; spaced look-back
4. AI chatbot	On-demand Socratic tutor via a server proxy.	Cognitive apprenticeship; optional

6. Pedagogy and learning design

Each topic follows a Bruner spiral packed into ten engine steps. Flagship missions (Force Fighter, Electrical Basics, ICT Fundamentals Mission 1, Chemistry Lab, and others) use an explicit Opening plus four Enactive/Iconic/Symbolic loops and a recap map. All finished missions gate Mastery behind an 80% fluency drill. Difficulty rises inside a topic; it does not jump to 'level 10 equals the hardest chapter of the whole subject.'

Mission cycle (engine pedagogy): Recall → Predict → Act → Feedback → 80% fluency → Mastery.

The 80% fluency gate is a product rule, not a slogan. Mastery stays locked until the drill score is earned. That is the opposite of 'time watched equals progress.' Flow Autopilot today is streak-based scaffolding (honest limit: not a full adaptive engine).

7. Product features

- 28 live catalog titles with a shared hub, coach, chips, and save format.
- Bilingual UI (EN/BN) including Specimen Lab labels.
- Digital mission books with linked terms that can open the tutor.
- Unlock-books demo toggle for juries who need to browse books quickly.
- Optional landing login (Laravel Sanctum) without blocking offline play.
- 3D Specimen Lab: orbit, numbered pins, curved callout to the part name and explanation.
- Pin editor for placing labels on models; placements can be exported as JSON.
- Secure Groq proxy with a chain of free chat models and a local fallback script.

8. Curriculum coverage

Track	Titles
Core sciences (5)	Force Fighter, Chemistry Lab, Bio Explorer, Math Quest, Eco Guardian
CS and tech (10)	ICT, Web, Backend, Database/SQL, Networking, Cyber Shield, OS & Hardware, AI Lab, ML Lab, Data Science
Engineering (5)	Electrical, Mechanical, Civil, Electronics & Robotics, Green Tech
Advanced science (4)	Astronomy & Space, Geology & Earth, Human Anatomy, Genetics & Biotech
Math extended (4)	Statistics, Geometry & Trig, Calculus, Discrete Math

9. Technology and architecture

Layer	Choice	Why this, not the alternative
Gameplay	HTML5 Canvas 2D + vanilla JS	No Unity/WebGL download; readable missions on low-end devices
Platform	Custom engine/ (hub, persist, pedagogy)	One product, not iframes inside an LMS
Landing	HTML/CSS/JS	Fast catalog, bilingual copy
AI tutor	Python groq_proxy.py + Groq API	Key never in the browser; games work if chat is down
Auth/admin	Laravel + Sanctum + Filament (optional)	School-owned admin; SQLite or MySQL
Progress	localStorage save-v2	Instant restore on flaky school networks
3D lab	Three.js (WebGL) Specimen Lab	Spatial inspection without forcing 3D onto every lesson

Architecture (learner in the center)

Learner → Landing + Mission Hub → Shared engine (UI, pedagogy, persistence) → Subject package (boot, missions, Canvas scenes) → localStorage. Side doors: Groq tutor proxy; optional Sanctum API + Filament.

10. Why missions are 2D and 3D is a lab

Graded missions need a stable camera, readable chips, and a coach that never hides behind a mesh. 3D everywhere would raise load time, asset size, and QA cost across 28 subjects. 3D none would waste the chance to inspect a cell, a heart, or an engine in space.

	Canvas 2D missions	3D Specimen Lab
Purpose	Structured outcomes, fluency, books	Inspect a model; pin parts; read a callout
Success	Pass the spiral; 80% drill	Find the organelle / part and name it
Device load	Must stay light on every title	Acceptable as an optional lab
Authoring	Fast scene + chips + coach	GLB + pins; not 280 missions of 3D

11. AI tutor (optional, never blocking)

Book terms open a chat that posts to /api/chat. The Python proxy keeps GROQ_API_KEY on the server. Groq retired llama-3.3-70b-versatile for free/developer use (16 August 2026). The proxy now walks a fallback chain of live free Groq text models: openai/gpt-oss-20b, openai/gpt-oss-120b, qwen/qwen3.6-27b, groq/compound-mini, groq/compound, openai/gpt-oss-safeguard-20b, allam-2-7b. Rate limits or a dead ID move to the next model. If Groq is blocked, a local tutor script still answers so class is not stuck.

- First tap on a word: full picture (5-7 sentences) and a fixed opt-in question.
- Follow-up turns: deeper Socratic question about the idea just taught.
- Speech, TTS, and prompt-guard models are not used for this chat path.

12. Comparison with typical EdTech

Dimension	Typical EdTech	GyanQuest
Engagement	Watch + MCQ	Act in a 2D mission
Progress	Time watched / completion %	80% fluency lock before Mastery
Access	Cloud required	Offline-first; chat is extra
3D	3D everywhere or none	3D Specimen Lab only, missions stay 2D
Language	English UI, Bangla as subtitle	Two strict language modes
Subjects	One app per topic	One engine, 28 titles

13. Methodology

- Map NCTB-adjacent school topics to 10-mission shells per subject.
- Design the Force Fighter spiral (hook to mastery) as the template for every title.
- Build Canvas scenes and persist (save-v2); QA uniqueness of activities where claimed.
- Author bilingual copy; keep Bangla in Noto Sans Bengali on the web client.
- Isolate 3D in Specimen Lab with pin JSON that teachers can correct.
- Keep Groq behind a proxy; test local fallback when the network fails.
- Iterate from playtests: coach clarity, pin placement, dark-theme consistency.

14. Results and current status

As of August 2026 the landing catalog lists 28 live titles. Sixteen games pass an internal uniqueness bar (LEVEL_QA.md). Fifty-two missions are finished to a playable standard. Remaining catalog mission slots (228 of 280) are structured stubs (same hub, coming-soon cards), which is a capacity choice, not a hidden crash. Recent remodels include ICT Fundamentals Mission 1 (Bits: Inside the Machine's Kitchen): CPU, RAM, and storage through a kitchen metaphor with live switch-flipping and counter-vs-pantry interactions. Specimen Lab ships with a catalog of classroom models and baked pin positions for animal cell, plant cell, heart, kidney, and city, plus an editor for the rest.

15. Impact

Impact equals converting school topics into mastery-gated, bilingual, offline missions at the cost of a browser, not a lab. Learners move from watching to enacting. Teachers get one pattern instead of 28 tools. Communities with shared phones are not locked out by a 3D-only download.

Alignment	How GyanQuest contributes
Quality education (SDG 4)	Active missions, fluency gate, Bangla/English parity
Industry and innovation (SDG 9)	Browser STEM lab; optional self-hosted admin
Reduced inequalities (SDG 10)	Offline-first; low device bar; no paid 3D runtime

16. Limitations and ethics

- Flow Autopilot is streak-based; it is not a full adaptive difficulty model.
- Mission 2+ depth still varies by title; stub slots remain on the hub until authored.
- 3D models come from Sketchfab-style classroom assets; credits stay on the lab sheet.
- AI answers can be wrong; the canvas and book remain the source of truth.
- No biometric or unnecessary personal data is required for core play.

17. Roadmap

Horizon	Work
Now	28 titles, 52 playable missions, fluency gate, Specimen Lab, Groq fallback chain, optional Laravel
Next	Teacher dashboard on Filament; deeper uniqueness on remaining titles; more baked 3D pins
Later	Limited 3D beats inside selected lessons; richer adaptive scaffolding

18. Conclusion

Team Alpha submits GyanQuest as a complete, demonstrable learning system: not a slide deck about gamification, but a browser lab with a pedagogy loop, a fluency lock, bilingual copy, an honest 2D/3D split, and an AI tutor that cannot hold the class hostage. We ask the jury to play a mission, pin a cell, and switch language. That is the product.

19. References

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